

cocotb DUT handle

Triggers tell you when to act

Path walk and dot value

- Dut is the hierarchy root the simulator hands your test
- Attribute dots walk instances
- Reading dot value peeks; assigning dot value pokes
- A missing name raises `AttributeError`, there is no silent `None` for a bad path
- Deeper nests like `dut dot core dot regs dot r zero` are the same idea

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

cocotb DUT handle

Sketch hierarchical **dut.path** access from Python: resolve a signal, then **peek** / **poke** **.value**. Starter: `dut.uart.txd` resolves — value 1.

Starter example: `dut.uart.txd` resolves — peek `.value` is 1.

Load starter example

Challenges 0 / 22 cleared

Quiz: **dut**: In a cocotb test, dut is...

☐ the hierarchy root handle into the simulated design

☐ only the Makefile MODULE name

☐ a replacement for RisingEdge

☐ the synthesis netlist file

Show hint

Check

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Idle

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Core ideas

dut

Hierarchy root handle into the design.

path

Dots walk instances: `dut.uart.txd`.

.value

Peek / poke the signal payload.

miss

Bad path → `AttributeError` (resolve FAIL).

Browser lab starter

learn_cocotb — cocotb dut handle

Real cocotb practice

- In the real cocotb track, restate the handle idea on paper before you touch a simulator
- Draw a tiny hierarchy: top, uart block, txd leaf
- Write one peek line and one poke line using dot value
- A ghost path that fails
- This module is handle literacy, not a full live UART simulation yet

Pitfalls to watch

- Watch for inventing hierarchy names that are not in the netlist
- A common trap is forgetting dot value; the handle is not the bit by itself
- Another miss is treating a successful resolve as correct stimulus
- The browser tree is literacy

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and try one missing-path challenge
- Explain path walk and dot value in your own words, then sketch one poke and one peek
- When you are ready